

# **Machine Learning vs Deep Learning**

## **Introduction**

Machine learning refers to large area of AI where algorithms are trained with the data and let them make decisions or predictions automatically rather than the specific programmed instructions. In other words, they consume input data to learn and improve results. Hence conventional machine learning needs human interaction for feature engineering, which involves manual selection of relevant features for the data in the algorithms such as decision trees or logistic regression

Deep learning is a one of the sub-categories of machine learning domain and consists of seamless structured neural networks. In contrast with conventional machine learning, deep learning models can extract the features by themselves and learn the data step by step using one layer of data at the time.

Both the machine learning (ML) and deep learning (DL) are parts of the area of artificial intelligence. The main point of difference lies in the way these processes handle information as well as how they onboard knowledge. With a view to this, the following report analyses both House Price Prediction (a Machine-Learning task) and Self-Driving cars (a Deep Learning task).

## **Case Study 1: Traditional Machine Learning and House Price Prediction**

House price estimation is based on structured data in which such features as surface area, number of rooms, and geographical position can be accounted for easily. Traditional ML with the help of such algorithms as linear regression is suitable for the task. What makes the models based on ML relevant is that the features have direct linear dependence with the price. This makes it

possible for ML-based models to be understandable for stakeholders who can see that change in square area directly influences the price. As an advantage, classical ML approaches do not have high computational costs and are efficient when working with small, structured datasets known to be useful.

When it comes to house price prediction, the deep learning approach is inappropriate. Deep learning entails using deep neural networks, which may cause overfitting whereby the algorithm interprets random fluctuations in data as valid signals. This is because deep learning approach aims at finding nonlinear hidden features of high dimensional data making it overly complex for this case.

## **Case Study 2: Deep Learning for Autonomous Driving**

Autonomous driving is reliant on processing massive flows of unstructured data, which include high-resolution imagery, LiDAR readings, and radar signals. Deep learning, particularly Convolutional Neural Networks (CNNs) is the only applicable method because it automates feature extraction. The algorithm learns how to recognize important visual formations including edges, textures, and finally complicated objects such as people or traffic signs without human input. The capacity of deep neural networks to deal with high-dimensional data in real time becomes especially crucial for navigating the unpredictable character of real-world driving.

Traditional machine learning is not appropriate for self-navigation at all. Traditional algorithms engage human-defined feature selection methods requiring human patience when it comes to autopilot situations in a city street, where it is impossible to prepare codes of every unpredictable case (e.g., abstract light effects, climate-related visibility issues, or unusual reactions

of pedestrians). As a result, any traditional machine learning approach could be potentially utterly inefficient in practical terms, unable to adapt itself to new surroundings and unknown obstacles.

## **Conclusion**

In summation, choosing the right AI approach depends on the type of data and complexity of the task at hand. Shallow machine learning remains the best choice for tasks that have structured data and can make effective use of human expertise for feature engineering, but deep learning is the tool to use for complex unstructured tasks where the data has high dimensionality as feature engineering can't be done manually.

The methodology of Artificial Intelligence must consider both the data characteristics and the complexity of a task. Traditional Machine Learning is applied in cases of structured problems, where human skills can provide feature engineering of the problem, while deep learning is necessary when the task is complex due to unstructured data and its high dimensionality, not making feature engineering humanly possible. It is important to align the methodology with the domain to ensure efficiency and performance of the task. The line between these approaches is becoming more blurred as the field is evolving. Hybrid systems and multi-modal learning emerge to allow the incorporation of deep learning components into traditional statistical methods for increased stability and understanding.

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